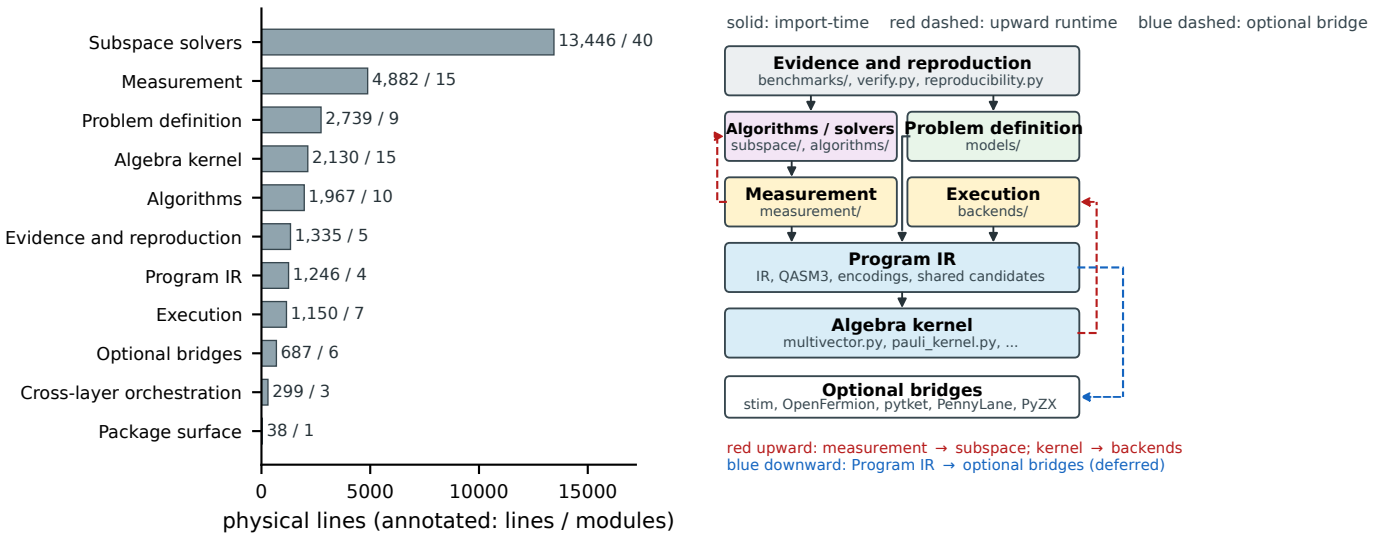
(a) Declared-layer size
Subspace solvers
13,446 / 40
Measurement
4,882 / 15
Problem definition
2,739 / 9
Algebra kernel
2,130 / 15
Algorithms
1,967 / 10
Evidence and reproduction
1,335 / 5
Program IR
1,246 / 4
Execution
1,150 / 7
Optional bridges
687 / 6
Cross-layer orchestration
299 / 3
Package surface
38 / 1
0
5000
10000
15000
physical lines (annotated: lines / modules)
(b) Selected dependency edges (partial)
solid: import-time
red dashed: upward runtime
blue dashed: optional bridge
Evidence and reproduction
benchmarks/, verify.py, reproducibility.py
Algorithms / solvers
subspace/, algorithms/
Problem definition
models/
Measurement
measurement/
Execution
backends/
Program IR
IR, QASM3, encodings, shared candidates
Algebra kernel
multivector.py, pauli_kernel.py, ...
Optional bridges
stim, OpenFermion, pytket, PennyLane, PyZX
red upward: measurement → subspace; kernel → backends
blue downward: Program IR → optional bridges (deferred)